

The tempo of automation: how job content changes in response to technological change

Working paper — companion to “Technology fields, automation and task reinstatement”

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Abstract

The companion paper closes the spatial task-based model as a static equilibrium and reads its consequences as comparative statics between two rest points. This paper turns the same model into laws of motion and asks what the static account cannot: not where reinstated work ends up in equilibrium, but how the transition unfolds in time, and how its outcome depends on the speed of the shock. The technology matures gradually — a logistic diffusion over a calendar window — so that displacement, and the new tasks it seeds on the boundary of the technology field, are released as a flow rather than at once. The single new object the dynamics require is a stock of *unbound* task mass: the seeded work that has been created but not yet attached to any occupation, the third fate of new tasks that the static model creates and discards at every evaluation. Its trajectory $U(t)$ is the transitional mismatch between destruction and creation, and it is the model’s central dynamic observable. Binding draws this stock down: new work is attracted to the occupations whose held capabilities best match it, but each occupation can absorb only at a rate limited by its size, so a saturated best match leaves a residue that cascades to the next-best match. The outcome is governed by a single ratio, the tempo of the shock against the tempo of redistribution. When redistribution keeps pace, reinstated work is bound by the high-skill specialists whose capabilities fit the frontier; when the shock outruns redistribution, small best-matched occupations cannot absorb it and the mass is taken instead by the large, lower-skilled occupations that can, a forced downgrading. The same technology and the same field thus yield materially different distributions of work depending only on the tempo: the final allocations of the slow and fast regimes correlate at roughly 0.3. This path-dependence of the destination, and not only of the transitional cost, is invisible to the equilibrium framework, whose fixed point sees only the endpoint. The realisation is illustrative, inheriting the calibration of the companion paper.

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1 Introduction

A technology does not arrive all at once. It emerges, diffuses, and matures over years, and while it does, the labour market is already responding: displacement, the creation of new tasks, and the reallocation of workers run in parallel, not in sequence. The static task-based framework cannot see this. It computes an equilibrium — a state that satisfies a fixed-point condition — and reads the consequences of a technology as the difference between two such rest points (**acemoglu2018race**; **acemoglu2019automation**). The intermediate path, and with it the cost of the transition and any dependence of the outcome on the speed of the shock, is by construction absent. Yet the transition is where the human consequences of automation are felt, and the central claim of this paper is that the speed of the shock relative to the speed of adjustment determines not merely how costly the transition is but *who ends up doing the reinstated work*.

The companion paper (**storck2026fields**) develops the spatial generalisation of the task-based framework on which we build. Occupations are bundles of tasks in a two-dimensional task space; the market price of skill is a directional field over that space; a technology is a Gaussian augmentation field; automation removes the high-priced tasks of a bundle; and reinstatement seeds new tasks on the gradient boundary of the technology field, which survive as human work only where labour remains the cheaper producer. That paper closes the worker layer as a static equilibrium and states explicitly that the adjustment dynamics are left for later work, and that the rise-and-fall of employment which **bessen2019demand** documents over time appears in it only as a cross-section over the demand elasticity, not as a trajectory. This paper is that later work.

We make one structural addition and otherwise reuse the companion model unchanged. The static model creates new task mass on the boundary ring and partitions it into three fates — captured by capital, bound to an occupation, and unbound — but it has no home for the unbound part: it is created and discarded at every evaluation. The dynamic model gives it a home, a stock $U(\mathbf{r}, t)$ of unbound task mass that fills as the technology matures and drains as occupations bind it. This stock is the only genuinely new object; population, price, displacement, seeding, and the bundle operator are the companion model's, now read as rates rather than as a solved equilibrium. The trajectory of the stock, $U(t)$, is the transitional mismatch between destruction and creation, and it is the first observable the static model cannot produce.

The second, and the paper's central result, concerns binding. Unbound work is attracted to the occupations whose held capabilities best match it, but an occupation can take on new work only at a rate limited by its size, so the best match, if small, saturates and leaves a residue that is offered to the next-best match. The realised allocation therefore depends on a race: the tempo at which the shock seeds new work against the tempo at which occupations can bind it. We show that this ratio governs not only the height and duration

of the transitional mismatch $U(t)$ but the final destination of the work. A gradual shock — redistribution faster than the shock — lets the best-matched occupations keep pace, and the reinstated work is bound by the high-skill specialists whose capabilities fit the frontier. A shock that outruns redistribution forces the work onto the large, lower-skilled occupations that can absorb it fastest, a downgrading that the gradual transition avoids. The two regimes are run through the same technology and the same field; their final allocations of reinstated work correlate at roughly 0.3. The destination, not just the transitional cost, is path-dependent, and the equilibrium framework sees none of it.

[TODO: Position against the dynamic adjustment literature once the frame is fixed (e.g. mismatch and reallocation after sectoral shocks); decide how much to engage with the equilibrium-with-adjustment-cost tradition versus presenting the result as a property the spatial geometry makes visible. Keep the empirical-illustration discipline of the companion paper: no causal claim is made from the illustrative calibration.]

2 The static foundation

We recall only what the dynamics require; the construction, calibration, and interpretation are in the companion paper ([storck2026fields](#)).

Tasks lie on the unit disk in polar coordinates $\mathbf{r} = (\xi, \chi)$, with ξ a domain orientation and χ a depth. The market price of skill is a directional field

$$\ln \Pi(\mathbf{r}) = m_0 + m_1 \cos \xi + m_2 \sin \xi + \chi (m_3 + m_4 \cos \xi + m_5 \sin \xi), \quad (1)$$

high in the socio-cognitive directions and low in the service and manual ones. A technology is a Gaussian field $\varphi_K(\mathbf{r}) = A_K \exp(-\frac{1}{2}\|\mathbf{r} - \mathbf{p}_K\|^2/z_K^2)$, with maturity A_K , centre $\mathbf{p}_K$, and reach z_K . Capital bears a task only where it is the cheaper producer, so the operated share is

$$a(\mathbf{r}) = \sigma\left((s_K \varphi_K - R/\Pi)/\tau\right), \quad (2)$$

which crosses first where the price Π is high: automation takes the dearest work. Each occupation o is a bundle $b_o(\mathbf{r})$ with centroid μ_o ; its displaced mass is $D_o = \int b_o a$, and the aggregate displacement is $\Delta\Gamma^D = \sum_o L_o D_o$.

Reinstatement creates new task mass $M = \gamma \Delta\Gamma^D$ on the gradient ring of the field, $\hat{g}(\mathbf{r}) = \|\nabla \varphi_K\| / \int \|\nabla \varphi_K\|$, which peaks at distance z_K from the centre. A seeded task survives as human work only where the operated share is low, through the survival gate $(1 - a)$: deep in the core, where capital dominates, the same seed is captured by capital. The surviving seed density is therefore $\hat{g}(1 - a)$, concentrated on the boundary crescent. An occupation can take a seeded task only where it already holds the priced capability the task requires; in the static model this readiness is the one-sided, v -weighted deficit gate

$e_o(\mathbf{r}) = \exp(-\delta_o(\mathbf{r})/\ell)$ with $\delta_o = \sum_k v_k \max(q_k - q_{o,k}, 0)$, and the attachment capacity of a location is $C(\mathbf{r}) = \sum_o L_o e_o$.

Two features of this foundation are load-bearing for what follows. First, displacement falls *within* occupations, on the interior margin $a(\mathbf{r})$, not between regions. Second, the survival gate places the surviving reinstatement on the low-priced side of the boundary: humans retain the work that is not worth automating, which in this price field is the cheaper, lower- Π work. The destination results below interact with this second feature, and we return to it in the discussion.

3 From equilibrium to motion

The static and dynamic models are not one model seen two ways; they are different models that answer different questions. The static model computes a state $L = F(L)$ without intervening time; its solver's intermediate steps are numerical artefacts and its damping is a convergence parameter, not a speed. The dynamic model writes laws of motion — differential equations in time — and lets the state move under them, with inertia and an actual elapse of time. That the dynamic state may approach a rest point is a consequence of the laws, not something computed, and in general it is never reached, because the target shifts while the technology matures.

System dynamics supplies the grammar: stocks change only through flows, and flows form loops. The geometry supplies the constitutive laws: every flow rate is a functional of the geometric fields. The technology maturity $A_K(t)$, the population $L_o(t)$, the unbound stock $U(\mathbf{r}, t)$, and the slow drift of occupational character are the stocks; of these, only U is new.

3.1 The unbound stock

In the static model the seeded mass M has three fates: a part $s a$ is captured by capital, a part $s(1 - a)\Phi$ binds to occupations, and a part $s(1 - a)(1 - \Phi)$ is unbound, where $s = M\hat{g}$ and Φ is the bound share. The unbound part performs no work and does not enter the density; it is created and discarded at each evaluation. The dynamic model gives it a stock,

$$\partial_t U(\mathbf{r}, t) = \dot{s}(\mathbf{r}, t) - \sum_o \iota_o(\mathbf{r}, t), \quad \dot{s}(\mathbf{r}, t) = \gamma [\partial_t \Delta \Gamma^D]_+ \hat{g}(\mathbf{r}) (1 - a(\mathbf{r}, t)), \quad (3)$$

where $\dot{s}$ is the seeding rate and ι_o the binding flow to occupation o (§3.3). The seeding follows the *rate* of displacement: new tasks are created while the technology matures, and seeding ceases once displacement plateaus. The unbound stock is the only structural addition; everything else is the companion model read as a rate.

A point of bookkeeping matters here. The dynamics do not impose global conservation of task mass, and should not: seeding is a genuine source (new tasks are created). What is conserved is the population, $\sum_o \dot{L}_o = 0$, and the disposition of created task mass among its three fates. The stock U closes the seeding ledger — it gives the unbound fate a carrier — not a spurious global conservation.

3.2 Population and maturation

The population moves toward the static reallocation target, but from the *current* distribution rather than a frozen baseline:

$$\dot{L}_o = \frac{1}{\theta_L} (T_o(L, t) - L_o), \quad T_o(L, t) = \sum_{o'} L_{o'}(t) P(o | o'; W(L, t)), \quad (4)$$

with P a row-stochastic origin–destination logit, so $\sum_o \dot{L}_o = 0$ exactly. The timescale θ_L is a calibrated speed in interpretable units; it replaces the static solver’s damping, inheriting nothing from it. First order is deliberate: it yields lags and gaps but no overshoot.

The technology matures as a logistic diffusion anchored in calendar time. With t in years and a shock window T_{shock} , the slope is set so that A_K rises from five to ninety-five per cent of its mature value over T_{shock} years, centred at $T_{\text{shock}}/2$:

$$A_K(t) = \frac{\bar{A}_K}{1 + \exp\left(- (5.88/T_{\text{shock}}) (t - T_{\text{shock}}/2)\right)}. \quad (5)$$

Anchoring the shock in calendar time makes θ_L and the binding timescale below calendar quantities, so that the outcome is governed by the *ratio* of the shock tempo T_{shock} to the redistribution tempo. We use $T_{\text{shock}} = 5$ years throughout as an illustrative diffusion window. **[TODO: Justify or sweep T_{shock} ; note that only the ratio to the redistribution tempo matters, so the choice is a normalisation of the time axis, not a free result.]**

3.3 Binding: match allocates, size limits the rate

The binding flow ι_o rests on a separation that proved load-bearing: *where* the unbound mass goes is set by match, *how fast* it binds is set by size. At each location the mass is attracted to the best match through a sharpened share, and each occupation binds up to a size-limited capacity:

$$t_o(\mathbf{r}) = U(\mathbf{r}) \frac{\text{FIT}_o(\mathbf{r})^\beta}{\sum_{o'} \text{FIT}_{o'}(\mathbf{r})^\beta}, \quad \text{FIT}_o(\mathbf{r}) = e_o^{\text{match}}(\mathbf{r}) e^{-\|\mathbf{r} - \mu_o\|/\rho}, \quad (6)$$

$$c_o = M_o/\theta_{\text{abs}}, \quad M_o = b_o^{\text{orig}} + r_o, \quad \iota_o(\mathbf{r}) = t_o(\mathbf{r}) \min\left(1, \frac{c_o}{\int t_o d\mathbf{r}}\right). \quad (7)$$

The claim t_o distributes the unbound mass by sharpened match: the exponent $\beta > 1$ lets the best match dominate the claim at each location. The capacity c_o scales with the occupation's task mass M_o , which grows endogenously with the reinstated mass r_o it has already bound. What a saturated occupation cannot staff this step — the fraction $1 - \min(1, c_o / \int t_o)$ — returns to U and is offered, at the next step, to the next-best match. Large occupations therefore bind faster in absolute terms but not faster relative to their size, and the residue cascades down the match ranking.

The readiness in FIT_o is a *two-sided* match distance e_o^{match} , penalising both under- and over-qualification, in contrast to the one-sided deficit gate e_o of §2, which is retained for the static capacity. The reason is that being able to do a task and becoming its home are different questions: an occupation far above a task's level can perform it but should not draw it in as its own work. **[TODO: State the two-sided distance explicitly and reconcile with the companion paper's one-sided gate, which is retained both for the static capacity and because its direction generates the cross-system mobility asymmetry of Paper 2.]**

The form was chosen against an alternative that conflates the two roles. A binding rate proportional to size times match, $\iota_o \propto M_o \text{FIT}_o$, lets size *multiply* match: in that form the absolute absorption correlates with occupation size at roughly 0.99 and with match quality at roughly 0.04 — size decides the destination and match governs only the relative growth. The match-allocated form with a size cap breaks this; the correlation of absorption with size falls to roughly 0.04, and the destination is governed by match, with size demoted from setting *where* the work goes to *how fast* it binds, which is its proper role. The sharpness β and the absorption timescale θ_{abs} are calibration handles, not worldviews; their economic content — whether absorptive capacity scales linearly or sub-linearly with size, and how sharply the best match wins — is settled by a sweep, and the central result below is itself such a sweep. **[TODO: Fix or bound β and θ_{abs} ; report the result's sensitivity to both. State the economic interpretation of each as a calibration target.]**

4 Results

We run a single technology field, calibrated to task-level AI exposure as in the companion paper, through the dynamic economy. The field matures over $T_{\text{shock}} = 5$ years; we vary the redistribution tempo θ (taking θ_L and θ_{abs} together) against this fixed shock. **[TODO: State the calibration cross-reference precisely and report the parameter table; confirm population conservation and the drain of U to zero as reproducibility checks.]**

4.1 The transitional mismatch

Figure 1 shows the shock and the mismatch it creates. The left panel is the logistic diffusion of (5), rising over the five-year window. The right panel is the unbound stock $U(t)$ for four redistribution tempos. When redistribution is faster than the shock, the mismatch is barely visible: the market binds the displaced work as fast as it is seeded. As redistribution slows, the mismatch hump grows — roughly twenty-fold from the fastest to the slowest tempo shown — and peaks later, within and just after the adoption window. In every case it drains to zero once diffusion saturates and binding catches up. There is no permanent mismatch in the model: all created work is eventually bound. The transitional cost lives entirely in the path — in the height and persistence of the hump — not in the endpoint, which is exactly what the static model, seeing only the endpoint, cannot register.

4.2 The destination depends on the tempo

The hump measures the cost of the transition; the same ratio also determines its outcome. Figure 2 shows where the reinstated work is bound in two regimes, run through the same technology and the same field. In the gradual regime, with redistribution faster than the shock, the best-matched occupations keep pace and absorb the work: the gainers are the high-skill specialists whose capabilities sit at the frontier of the field. In the congested regime, with redistribution much slower than the shock, the small best-matched occupations cannot bind fast enough — their size-limited capacity $c_o = M_o/\theta$ is small, the mass accumulates in U , and it is taken instead by the large, lower-skilled occupations that can absorb it quickly. The same work is downgraded.

The shift is not marginal. The final allocations of reinstated mass in the two regimes correlate at roughly 0.3: substantially different destinations from one technology and one field, differing only in the tempo of the shock relative to the market’s capacity to respond. This is the destination counterpart of the transitional hump, and it is a statement the equilibrium framework cannot make. The fixed point sees the endpoint of a frictionless reallocation; it cannot see that the endpoint the economy actually reaches depends on how fast it was forced to move.

[TODO: Add the per-occupation evidence behind the figure: the size-versus-match correlation in each regime, the quartile shares of absorption, and a short table of the leading gainers in each regime. Replace any relative-change ranking with absolute and spatial measures, since the relative measure is unbounded for small occupations.]

5 Discussion

The result is that the tempo of a technology shock, relative to the speed at which the labour market reallocates, is not merely a determinant of how painful the transition is but of its outcome. A gradual diffusion lets work find the occupations whose capabilities match it; a shock that outruns adjustment forces a downgrading that persists in the final allocation. The same field, the same displaced mass, and the same seeded reinstatement yield different societies depending on the pace. This is a genuinely dynamic statement, and it is the kind of statement the task-based equilibrium framework is not built to make: its rest point is the destination of a frictionless economy, and the friction here is time itself.

Two qualifications bound the claim, and both are inherited rather than introduced. The first is the survival gate. The companion model places surviving reinstatement on the low-priced side of the boundary, because a seeded task is retained by labour where labour is the cheaper producer at a *uniform* cost R , which is the low- Π , cheaper work. The reinstated work is therefore lower-priced to begin with, and the congested regime compounds this by routing it to the large low-skill occupations. Whether reinstatement *ought* to be lower-skilled is a question about the survival gate, not the binding law: in **acemoglu2019automation** reinstatement is new *complex* work in which labour has a comparative advantage, often higher-skilled, and capturing that would require the survival condition to turn on relative rather than absolute productivity — a varying human productivity field, high where the machine is weak, in place of a uniform R . That is a change to the static foundation, deferred here and flagged in the companion paper. The destination results are therefore conditional on the survival-gate economics, exactly as the companion paper’s incidence results are. **[TODO: Decide how far to take the comparative-advantage qualification here versus leaving it entirely to the foundation paper. At minimum, state plainly that the high- versus low-skill character of the gainers is conditional on the gate.]**

The second qualification is the binding calibration. The allocation depends on the sharpness β and the absorption timescale θ_{abs} , and the central result is presented as a sweep over the tempo precisely so that the dependence is exhibited rather than hidden. But the regime boundary — how slow redistribution must be, relative to the shock, for downgrading to set in — is a quantitative claim that rests on these handles, and a reader is owed their economic interpretation and the result’s sensitivity to them.

Several limitations remain. The realisation is illustrative, inheriting the companion paper’s calibration and its discipline: no causal claim is made from it. The unbound stock drains to zero, so the model has no permanent mismatch and no involuntary non-employment; the transitional cost is the hump, and a model with task decay or irreversible binding would carry a permanent imprint of the shock speed that this one does not. New tasks enter as added weight on existing locations rather than as new points on the disk.

And the relative-change measure of who gains is unbounded for small occupations, so the destination results should be read in absolute and spatial terms.

[TODO: Relate to the adjustment-cost and mismatch literatures once the framing is fixed; decide whether the permanent-imprint extension (task decay or forced downgrading that does not reverse) belongs here as a remark or in a further paper.]

6 Conclusion

Turning the spatial task-based model from an equilibrium into laws of motion requires one new object, a stock of unbound task mass, and yields two things the equilibrium cannot. The first is the transitional mismatch $U(t)$, the gap between destruction and creation as a transient that grows with the tempo of the shock and drains as the market catches up. The second, and the central result, is that the tempo of the shock relative to the speed of redistribution determines the destination of the reinstated work: a gradual shock matches it to high-skill specialists, a fast shock forces it onto large lower-skilled occupations. The same technology yields a different distribution of work depending only on how fast it arrives. The cost and the outcome of automation are, in this sense, properties of the path and not only of the endpoint.

[TODO: Bibliography keys to add/verify in references.bib: storck2026fields (the companion static paper, “Technology fields, automation and task reinstatement”, Storck & Andersson, 2026 — the foundation cited throughout); storck2026geometry (Paper 1, polar geometry, already in references.bib); acemoglu2018race, acemoglu2019automation, bessen2019demand (already needed by the companion paper). No new external references are strictly required for this draft; all machinery is inherited. Verify each against full text before submission.]

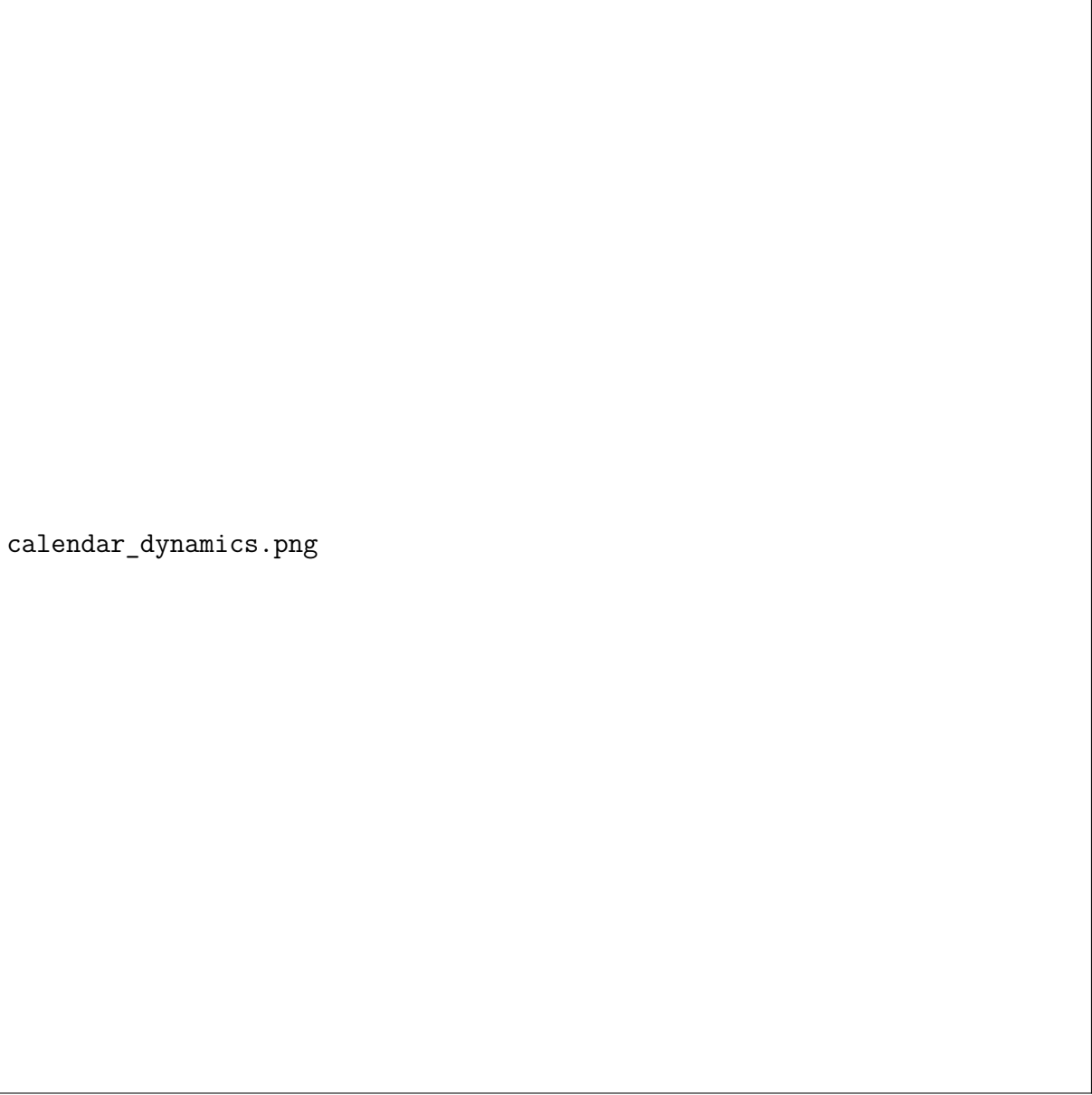
calendar_dynamics.png

Figure 1: Calendar-time dynamics. Left: the technology matures as a logistic diffusion over a five-year window. Right: the unbound stock $U(t)$, the transitional mismatch between destruction and creation, for four redistribution tempos θ against the fixed five-year shock. The mismatch hump grows and lingers as redistribution slows relative to the shock, but always drains to zero: the transitional cost is in the path, not the endpoint.



tempo_regimes.png

Figure 2: The destination of reinstated work depends on the tempo ratio. Both panels show the reinstated task mass gained by each occupation, over the survival-gated seeding field (grey) with the technology field dashed, for the same five-year shock. Left, a gradual transition: the best matches keep pace and high-skill specialists at the frontier absorb the work. Right, a congested transition: small best-matched occupations cannot keep up and the work is forced onto large, lower-skilled occupations. The two final allocations correlate at roughly 0.3.